

Mini Project

1: Inverter 2X:

1: LVS

The screenshot shows the LVS Comparison Results window for the cell Div_2x_inverter. The window is divided into two main sections: a table at the top and a summary section below.

Layout Cell / Type	Source Cell	Nets	Instances	Ports
Div_2x_inverter	Div_2x_inverter	4L, 4S	4L, 4S	4L, 4S

Below the table, the summary section shows the cell name and source name, both as Div_2x_inverter. It also displays the initial numbers of objects:

Ports:	Layout	Source	Component Type
4	4	4	
Nets:	4	4	
Instances:	4	3	MN (4 pins)
	4	2	MP (4 pins)

To the right of the window is a circuit diagram showing a 2x inverter layout with various components and connections.

2: DRC

The screenshot shows the DRC Check window for the cell Div_2x_inverter. The window is divided into two main sections: a table at the top and a list of errors below.

Check / Cell /	Results
Check DENSITY_PRINT_FILES	0

Below the table, a list of errors is shown, including:

- 00 DN 2L density
- 00 DN 2H 10 density
- 00 DN 2H CORE density
- 00 DN 3L density
- 00 DN 3H 10 density
- 00 DN 3H CORE density
- 00 DN 5 density
- 00 DN 1L density
- 00 DN 1H density
- 00 DN 2 density
- 00 DN 4 density
- 00 DN 1H density
- 00 DN 2 density
- 00 DN 4 density
- 00 DN 1H density
- 00 DN 2 density
- 00 DN 4 density
- 00 DN 1H density
- 00 DN 2 density
- 00 DN 4 density
- 00 DN 1H density
- 00 DN 2 density
- 00 DN 4 density
- 00 DN 1H density
- 00 DN 2 density
- 00 DN 4 density

To the right of the window is a circuit diagram showing a 2x inverter layout with various components and connections.

3: PEX

The screenshot shows the PEX Extraction Results window for the cell Div_2x_inverter. The window is divided into two main sections: a table at the top and a list of errors below.

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	DVDD	DVDD	27	1.75528E-16	1.83294E-16	3.58823E-16
2	DVDD	DVDD	8	1.70902E-17	2.09484E-16	3.06582E-16
3	OUT	OUT	11	1.03514E-16	3.64092E-16	4.73597E-16

To the right of the window is a circuit diagram showing a 2x inverter layout with various components and connections.

2: NAND 2X:

1: LVS

Navigator

Results

- Extraction Results
- Comparison Results

ERC

- ERC Results
- ERC Summary

Reports

- LVS Summary
- Extraction Report
- LVS Report

Rules

- Rules File

View

- Info
- Finder
- Schematics

Setup

- Options

Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
Div_2x_NAND	Div_2x_NAND	5L, 5S	3L, 3S	5L, 5S

Cell Div_2x_NAND Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

CORRECT #
#####

LAYOUT CELL NAME: Div_2x_NAND
SOURCE CELL NAME: Div_2x_NAND

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	5	5	
Nets:	6	6	
Instances:	4	4	NN (4 pins) NP (4 pins)

2: DRC

Filter: Show All

No Results Found

Check / Cell / Results

Check DENSITY_PRINT_FILES 0

OD DN 2L density
OD DN 2H 10 density
OD DN 2H CORE density
OD DN 3L density
OD DN 3H 10 density
OD DN 3H CORE density
P0 DN 2 density
P0 DN 3 density
M1 DN 1L density
M1 DN 1H density
M1 DN 2 density
M1 DN 4 density
M2 DN 1L density
M2 DN 1H density
M2 DN 2 density
M2 DN 4 density
M3 DN 1L density
M3 DN 1H density
M3 DN 2 density
M3 DN 4 density
M4 DN 1L density
M4 DN 1H density
M4 DN 2 density
M4 DN 4 density
M5 DN 1L density
M5 DN 1H density
M5 DN 2 density
M5 DN 4 density

Check DENSITY_PRINT_FILES

3: PEX

Navigator

Results

- Extraction Results
- Comparison Results
- Parasitics

Reports

- Extraction Report
- LVS Report
- Separate Properties

Rules

- Rules File

View

- Info
- Finder
- Schematics

Setup

- Options

Div_2x_NAND

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	DVSS	DVSS	23	1.83819E-16	1.75178E-16	3.58997E-16
2	DVDD	DVDD	16	1.27922E-16	3.34864E-16	4.62786E-16
3	OUT	OUT	12	4.69196E-17	5.02231E-16	5.49210E-16
4	A	NET1	2	6.97125E-18	1.31076E-16	1.38047E-16
5	A	A	11	7.16121E-17	3.48235E-16	4.19848E-16
6	B	B	11	8.36567E-17	2.83709E-16	3.67366E-16

Find Nets: Coupling to: All Nets Specified Nets

1: LVS

The screenshot shows the Cadence Virtuoso interface with the Comparison Results window open. The window displays the following information:

- Results:** Extraction Results, Comparison Results (selected).
- ERC:** ERC Results, ERC Summary.
- Reports:** LVS Summary, Extraction Report, LVS Report.
- Rules:** Rules File.
- View:** Info, Finder, Schematics.
- Setup:** Options.

The main window displays the Comparison Results for the cell **Div_4x_NAND**. The table shows the following data:

Layout Cell / Type	Source Cell	Nets	Instances	Ports
Div_4x_NAND B	Div_4x_NAND	5L, 5S	4L, 4S	5L, 5S

Below the table, the **Cell Div_4x_NAND Summary (Clean)** is displayed, showing the cell comparison results at the top level. The summary includes a diagram of the cell layout and the following information:

LAYOUT CELL NAME: Div_4x_NAND
SOURCE CELL NAME: Div_4x_NAND

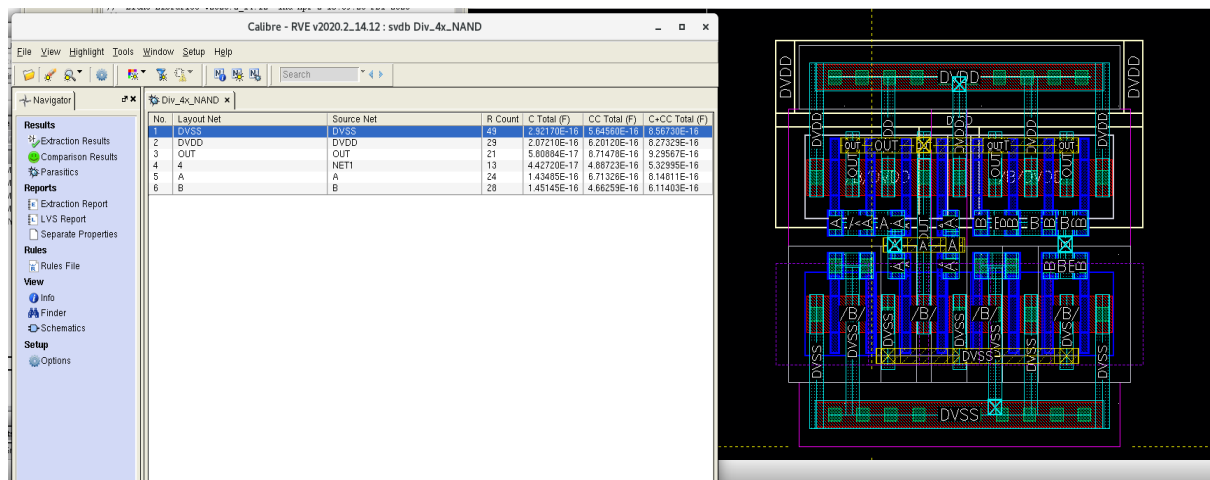
INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	5	5	
Nets:	6	6	
Instances:	8	4	* MN (4 pins)
	9	2	* MP (4 pins)

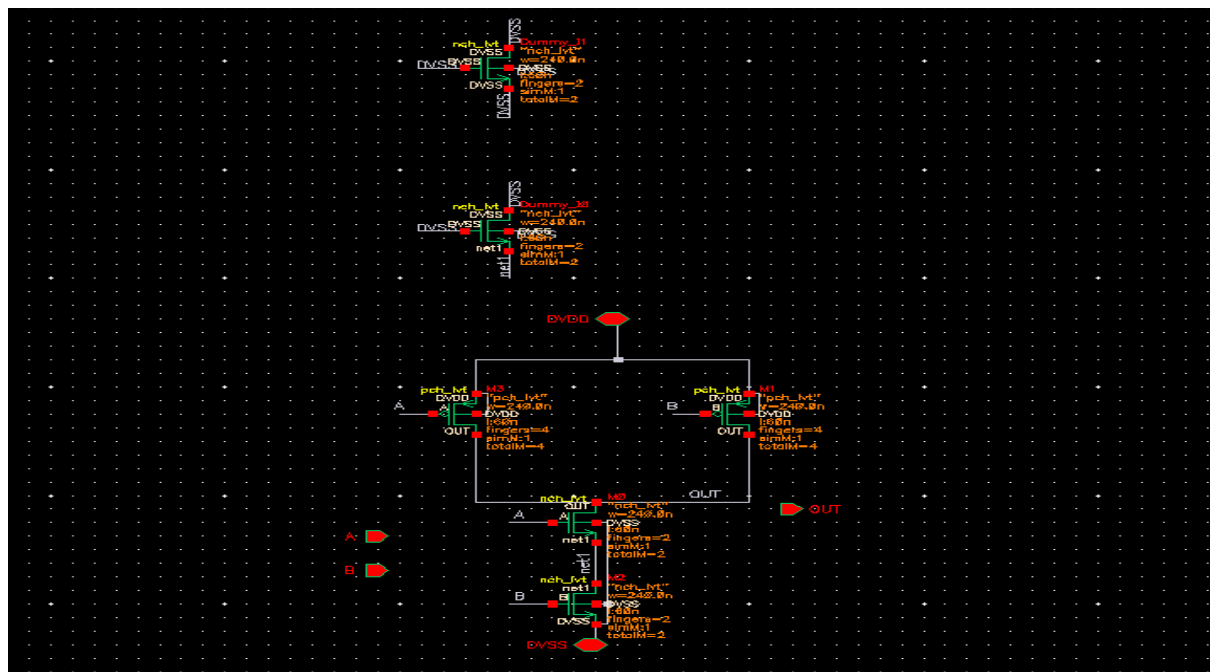
The right side of the image shows a detailed layout of the **Div_4x_NAND** cell, highlighting the internal structure and connections.

Figure 10 displays two screenshots from the Calibre RVE 2020.2.14.12 interface. The left screenshot shows the 'Check DENSITY_PRINT_FILES' dialog box, which lists various density files to be checked. The right screenshot shows a top-down view of a circuit board layout, highlighting various components and connections in different colors (green, blue, red, yellow) to represent different density levels or types.

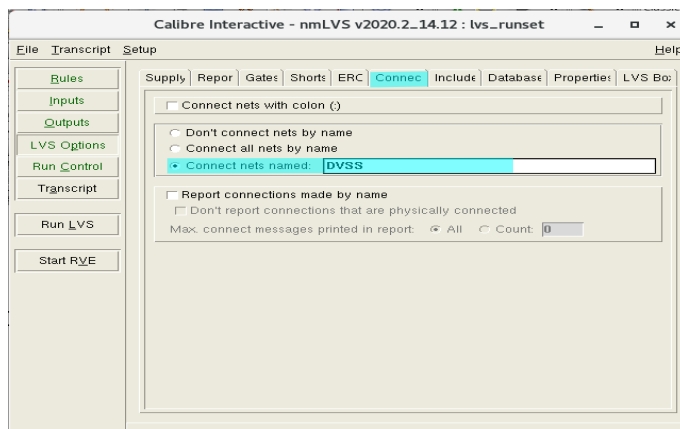
3: PEX



4: Schematic modification after adding dummies



Note (when I shared the DVDD and DVSS) I make this option in LVS & PEX to avoid giving error



4: Gated S-R Latch:

1: LVS

The screenshot shows the LVS extraction results for the cell `Div2_gated_SR_latch`. The interface includes a Navigator on the left with sections for Results, ERC, Reports, Rules, View, and Setup. The main window displays the following information:

Layout Cell / Type	Source Cell	Nets	Instances	Ports
Div2_gated_SR_latch	Div2_gated_SR_latch	9L, 9S	16L, 16S	6L, 6S

Cell `Div2_gated_SR_latch` Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

Warning: Unbalanced smashed mosfets were matched.

LAYOUT CELL NAME: Div2_gated_SR_latch
SOURCE CELL NAME: Div2_gated_SR_latch

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	6	6	
Nets:	13	13	
Instances:	32	32	* MN (4 pins)

The right side of the image shows a detailed layout view of the Gated S-R Latch circuit, featuring various transistors, capacitors, and interconnects.

2: DRC

The screenshot shows the DRC extraction results for the cell `Div2_gated_SR_latch`. The interface includes a Navigator on the left with sections for Results, ERC, Reports, Rules, View, and Setup. The main window displays the following information:

Calibre - RVE v2020.2.14.12 : Div2_gated_SR_latch.drc.results

Filter: Show All | No Results Found

Check / Cell / Results

Check DENSITY_PRINT_FILES 0

The right side of the image shows a detailed layout view of the Gated S-R Latch circuit, featuring various transistors, capacitors, and interconnects.

3: PEX

The screenshot shows the PEX extraction results for the cell `Div2_gated_SR_latch`. The interface includes a Navigator on the left with sections for Results, ERC, Reports, Rules, View, and Setup. The main window displays the following information:

Calibre - RVE v2020.2.14.12 : svdb Div2_gated_SR_latch

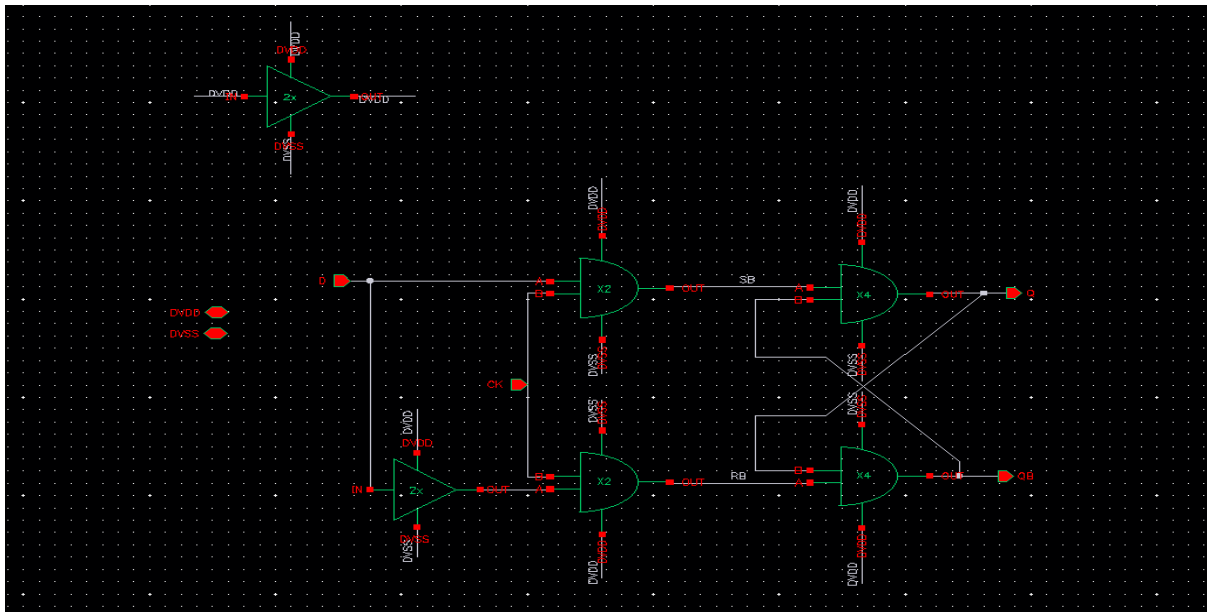
Filter: Show All | No Results Found

Check / Cell / Results

Check DENSITY_PRINT_FILES 0

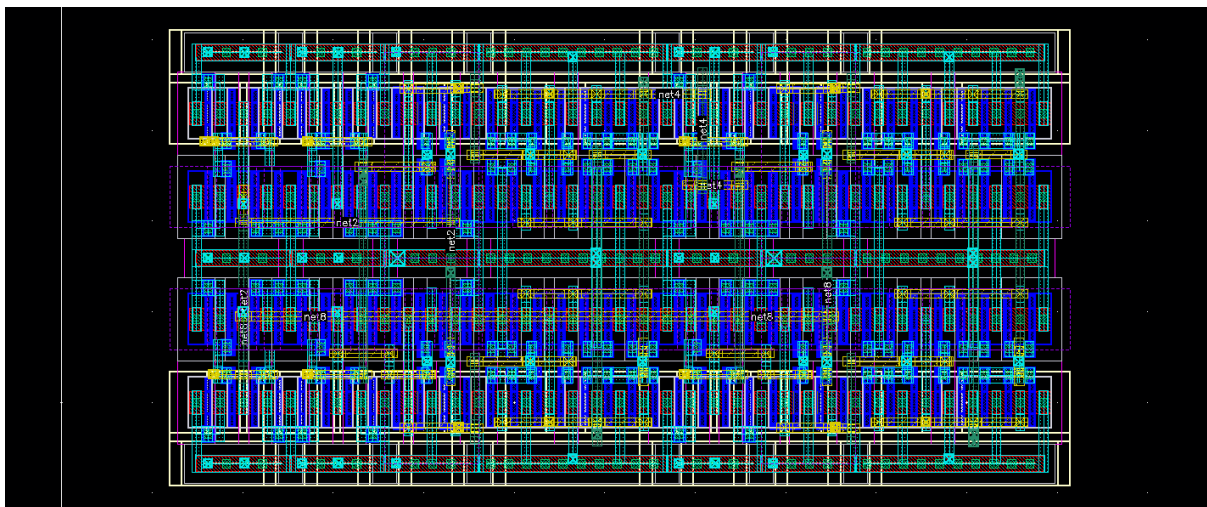
The right side of the image shows a detailed layout view of the Gated S-R Latch circuit, featuring various transistors, capacitors, and interconnects.

4: Schematic modification after adding dummies



5: Flip-Flop

Layout



1: LVS

The screenshot shows the LVS (Layout Versus Schematic) tool interface. The left pane displays the Results tree, including Extraction Results, Comparison Results, ERC Results, and Reports. The right pane shows the Cell Comparison Results for the Div2_FF cell. The results indicate that the layout and source cell are correct, with no mismatches found. The initial numbers of objects are also displayed.

Layout Cell Name	Source Cell Name	Nets	Instances	Ports
Div2_FF (B)	Div2_FF	16, 165	321, 325	6L, 6S

Cell Div2_FF Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

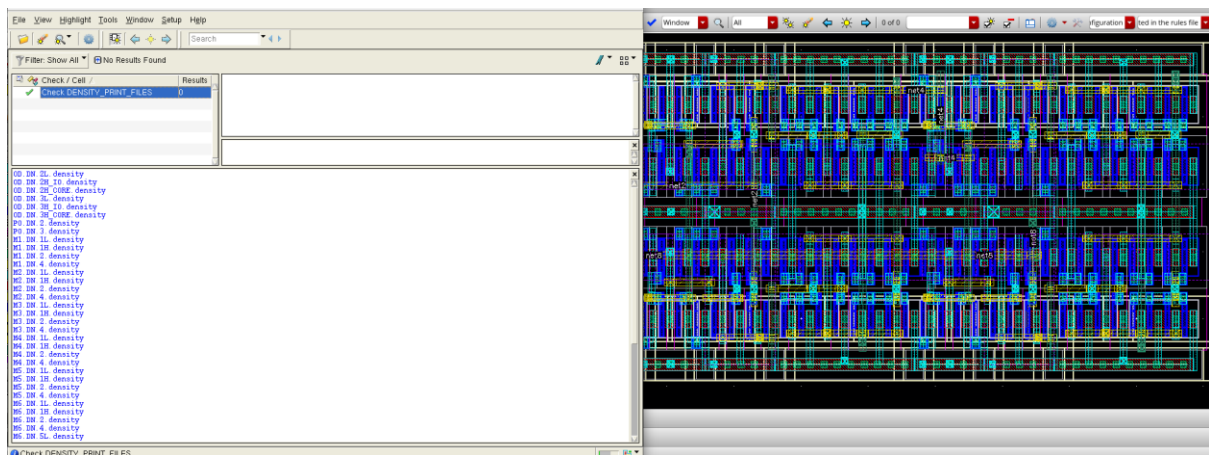
Warning: Unbalanced smashed xosfets were matched.

LAYOUT CELL NAME: Div2_FF
SOURCE CELL NAME: Div2_FF

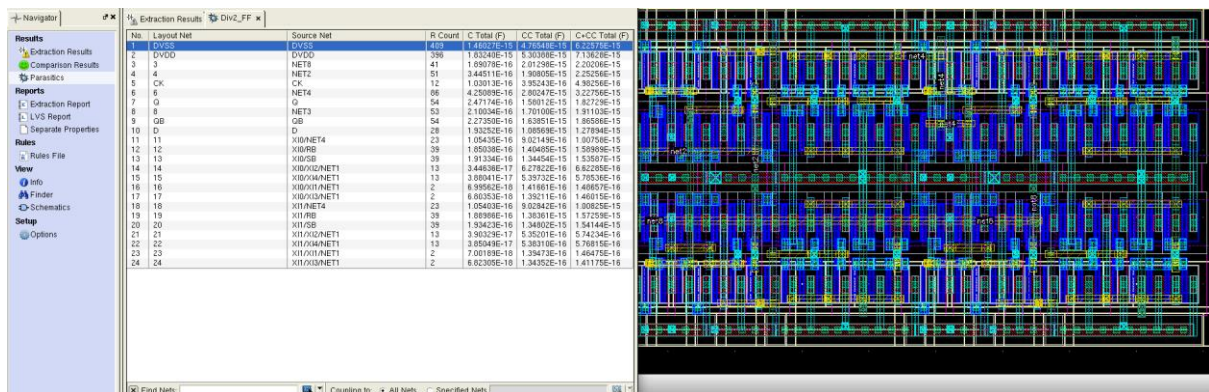
INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	6	6	
Nets:	34	34	
Instances:	72	50	XX (4 pins)

2: DRC



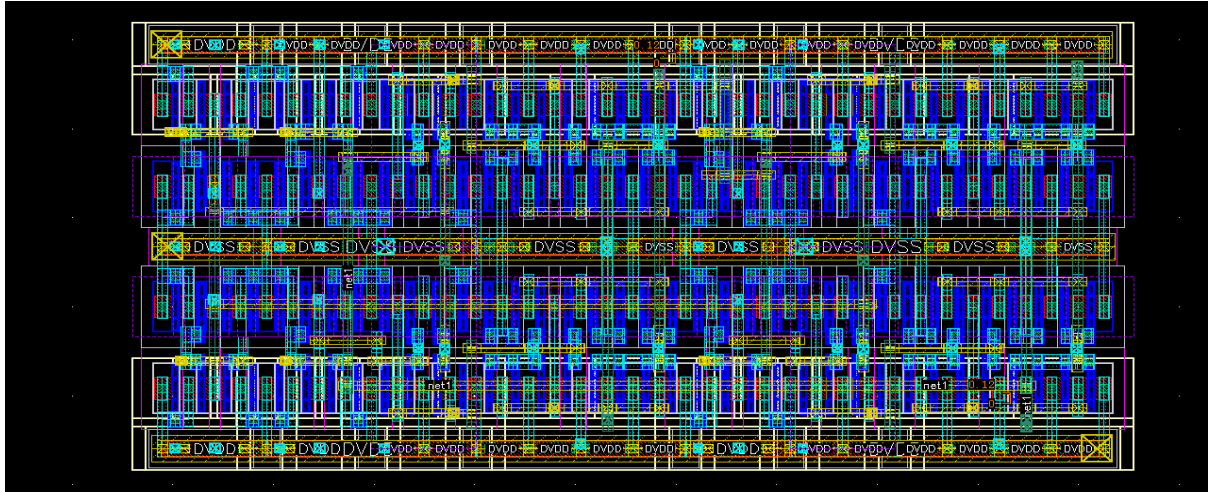
3: PEX



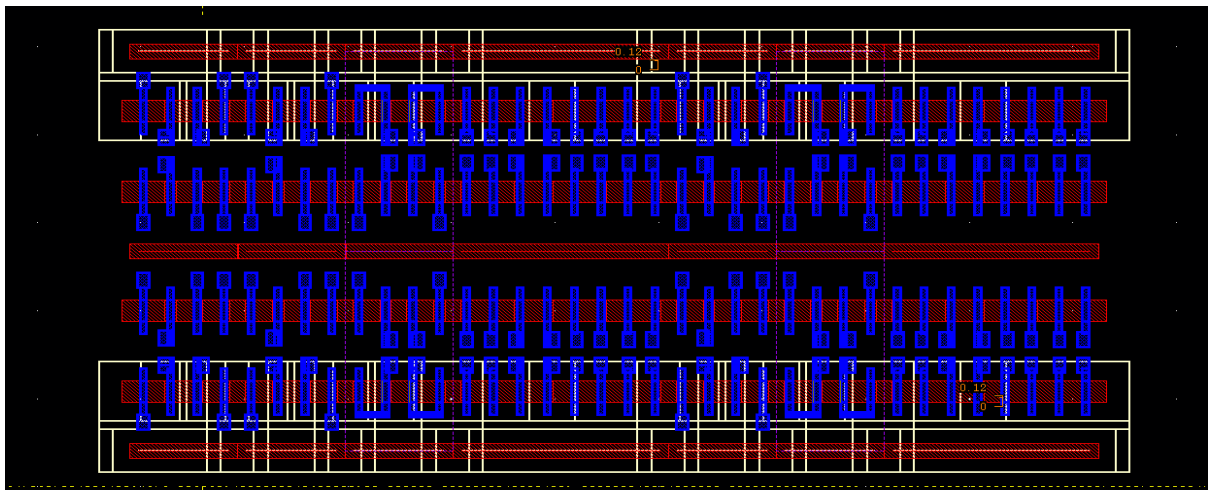
6: TOP Design

Layout

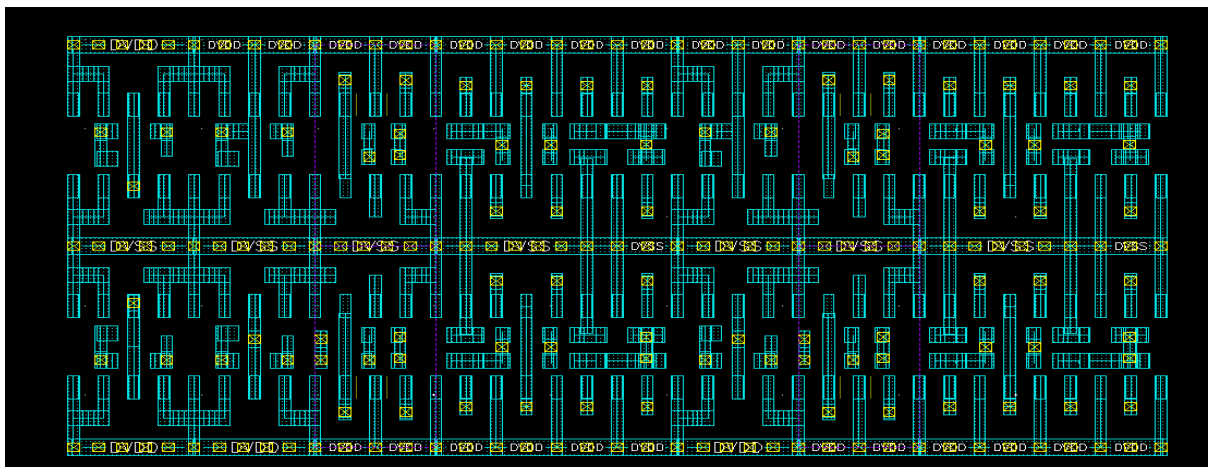
- One while showing all layers



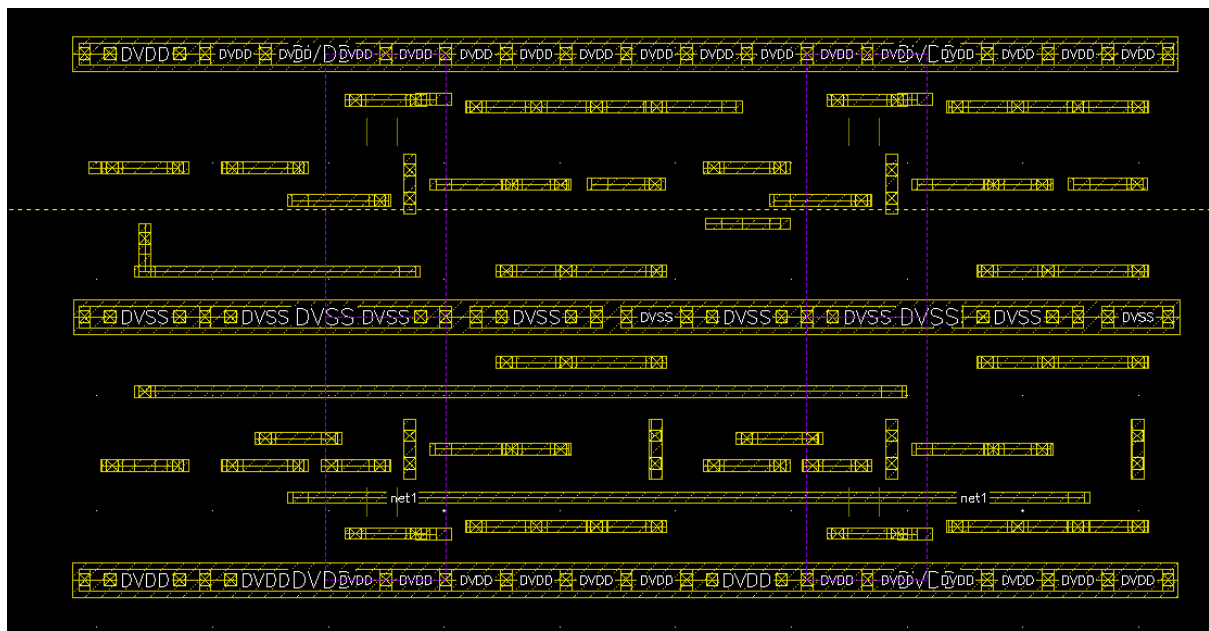
- One showing NW, OD, and PO together



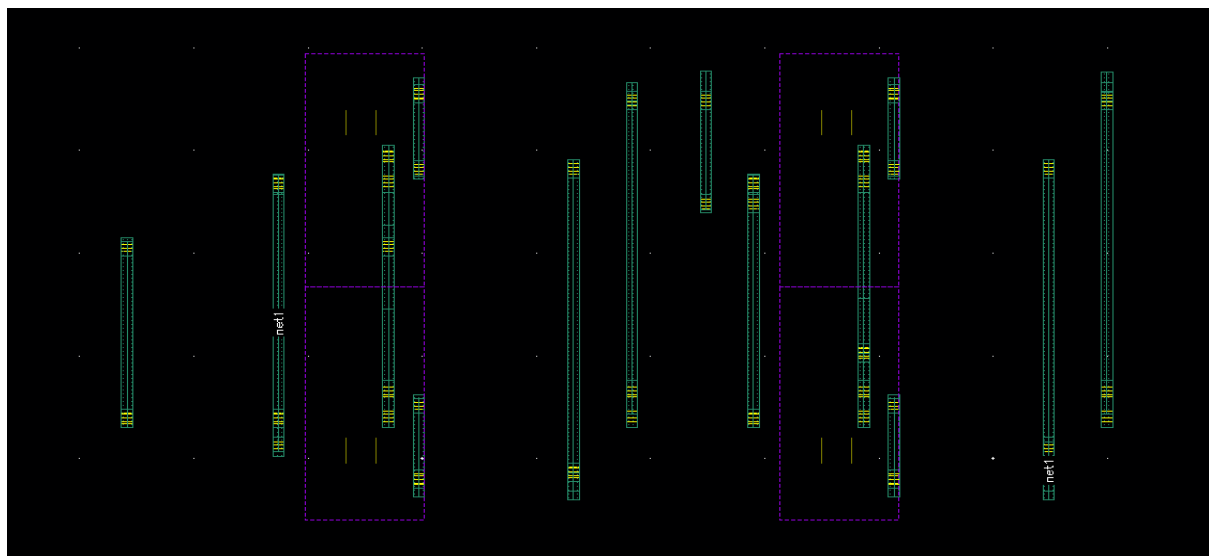
- A screenshot per each metal layer with the lower and upper VIA layers (M1)



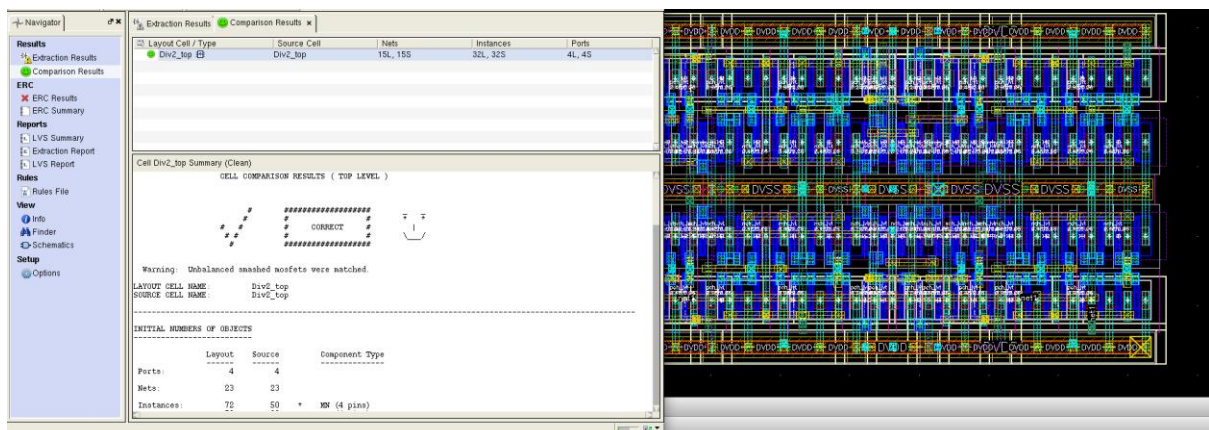
M2



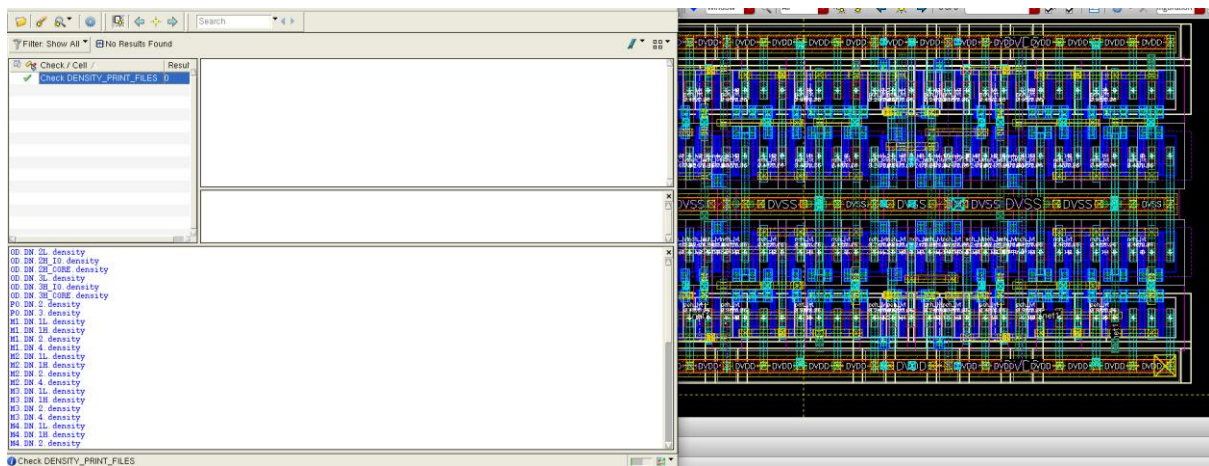
M3



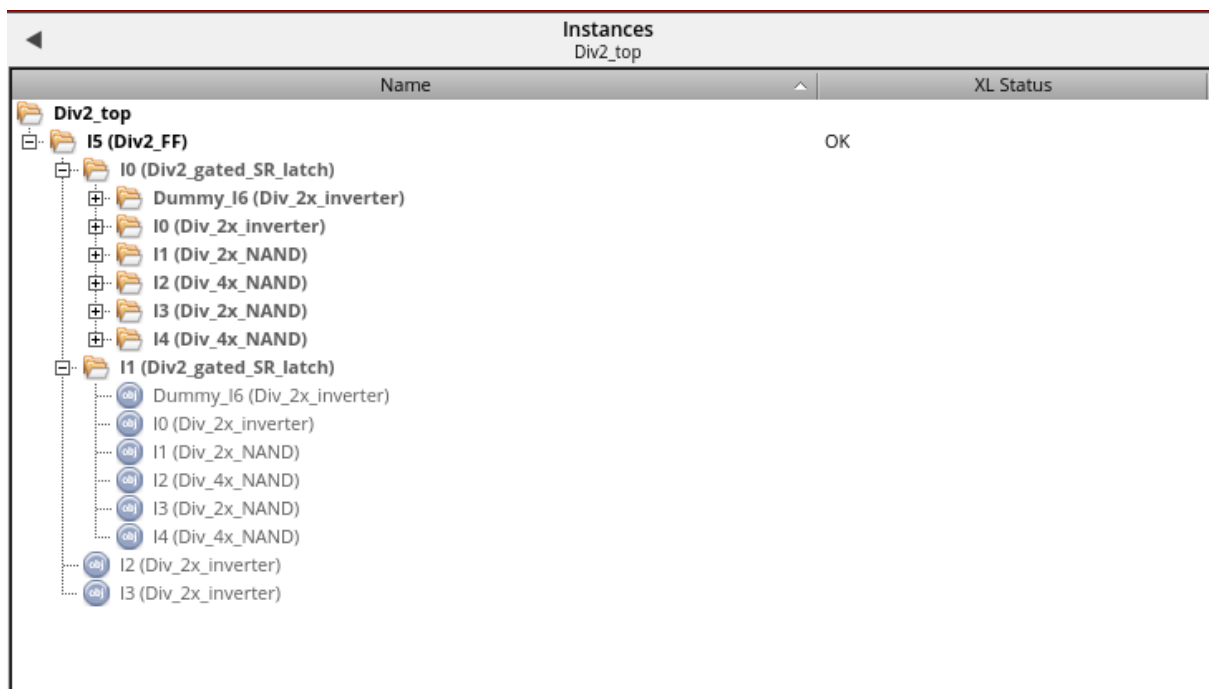
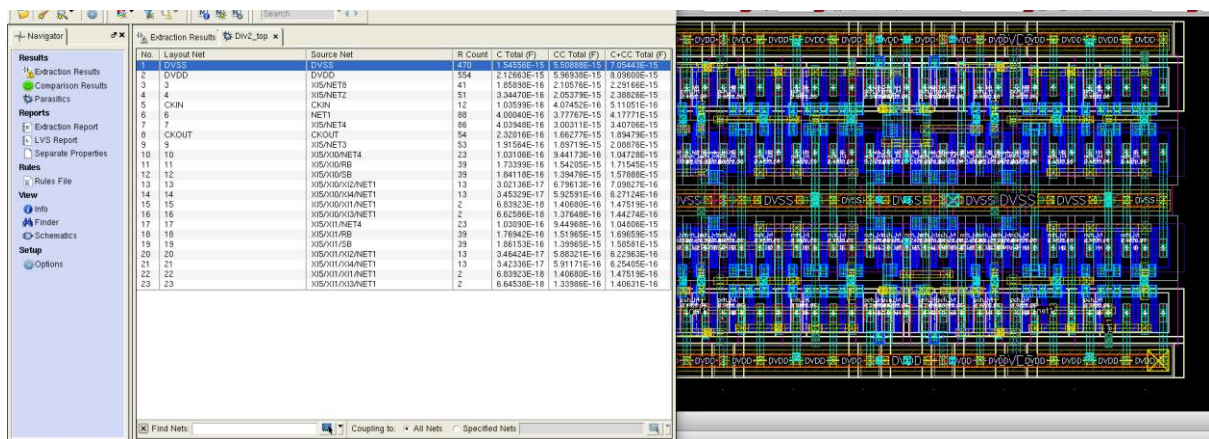
1: LVS



2: DRC



3: PEX (The total capacitance of the desired nets)



Nets		
Div2_top		
Name		XL Status
Div2_top		
I5 (Div2_FF)		OK
I0 (Div2_gated_SR_latch)		
I1 (Div2_gated_SR_latch)		
I2 (Div_2x_inverter)		
I3 (Div_2x_inverter)		
CK		
D		
DVDD		
DVSS		
net2		
net3		
net4		
net8		
Q		
QB		
CKIN		no markers
CKOUT		no markers
DVDD		no markers
DVSS		no markers
net1		no markers

Pins		
Div2_top		
Name		XL Status
Div2_top		
I5 (Div2_FF)		OK
I0 (Div2_gated_SR_latch)		
I1 (Div2_gated_SR_latch)		
I2 (Div_2x_inverter)		
I3 (Div_2x_inverter)		
CK:P_2		
D:P_3		
DVDD:P_4		
DVSS:P_5		
Q:P_0		
QB:P_1		
CKIN:P_1		OK
CKOUT:P_0		OK
DVDD:P_3		OK
DVSS:P_2		OK

REPORTS:

[LVS_DRC_PEX Reports - Google Drive](#)

Lab03 Library:

[Mini Project \(Lab03, 04\) - Google Drive](#)